

Numerical Investigation of High-Speed Flow over HIFiRE-1 Flight Research Vehicle



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B. Tech. in Aerospace Engineering

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Certificate

*This is to certify that the group project titled “**Numerical Investigation of High-Speed Flow over HIFiRE-1 Flight Research Vehicle**” is a bonafide record of the work carried out by **Mr. DEVENDRA SINGH DEORA, Ms. SUMAN S GOUDAR, Mr. VISHESH V.** bearing Reg. Nos. **19ETAS012011, 19ETAS012037, 19ETAS012045** Department of Automotive and Aeronautical Engineering - 2019 batch in partial fulfilment of requirements for the award of **B. Tech.** Degree of M. S. Ramaiah University of Applied Sciences.*

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Declaration

Numerical Investigation of High-Speed Flow over HIFiRE-1 Flight Research Vehicle

The project work is submitted in partial fulfilment of academic requirements for the award of **B. Tech.** Degree in the **Department of Automotive and Aeronautical Engineering of the Faculty of Engineering and Technology** of M. S. Ramaiah University of Applied Sciences. The project report submitted herewith is a result of our own work and in conformance to the guidelines on plagiarism as laid out in the University Student Handbook. All sections of the text and results which have been obtained from other sources are fully referenced. We understand that cheating and plagiarism constitute a breach of university regulations, hence this project report has been passed through plagiarism check and the report has been submitted to the supervisor.

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Summary

Humankind's desire to fly faster and the need for advanced military equipment to safeguard one's nation has fuelled the research in the field of hypersonic technology all over the world. To better understand the hypersonic flow physics, HIFiRE programme was developed as a joint collaboration between AFRL (U.S. Air Force Research Lab) and DSTO (Australian Defence Science and Technology Organization). The geometry of interest for this project work is also the HIFiRE-1 flight research vehicle, which was developed mainly to better understand the transition-to-turbulence and SBLI effects. This project however, focuses on the SBLI effects in the flare region.

Lack of adequate data in the high-speed flow regime fuels the research for development of new and improvement of existing technology and prediction models. This project provided a better understanding of the flow features over HIFiRE-1. All the simulations were carried out using the open source CFD software, SU2. Three different grid variants with coarse, medium and fine resolutions were considered to perform a grid convergence study. The discretisation of the geometry was carried out using Pointwise. To analyse and visualize the results as a part of the post-processing process, Tecplot software was utilised.

The axisymmetric simulation results obtained were in close accordance with the three-dimensional simulations reported in the literature. The flow features observed were also as expected based on the literature. The C_p and heat transfer data obtained is in reasonably good agreement with the literature. It is interesting to note that the SU2 simulation results are close to the experimental data whereas the ANSYS results are close to the literature simulation data.



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Nomenclature

C_p	Coefficient of pressure
C_h	Coefficient of heat transfer
Ma	Mach number
Re_u	Unit Reynolds number
Pr	Prandtl number
T_o	Total temperature
T_{aw}	Adiabatic wall temperature
r	Recovery factor
K	Turbulent kinetic energy
ω	Specific rate of dissipation
P	Static pressure
T	Static temperature
X	Axial position
t	Flight time



Abbreviation and Acronyms

AFRL	Air Force Research Laboratory
BLT	Boundary Layer Transition
DSTO	Defence Science and Technology Organization
HF1	HIFiRE-1
OMC	Optical Mass Capture
RANS	Reynolds Averaged Navier Stokes
SBLI	Shock Boundary Layer Interactions
SU2	Stanford University Unstructured

1. Introduction

A hypersonic flight is said to be achieved when a body moves through the atmosphere with speed greater than five times the speed of sound, in other words, the Mach number is greater than 5. A hypersonic missile can be optimised to a certain type of flight path such as cruise, ballistic or that of a boost-glide configuration. These missiles travel at high speeds reducing its time for response. Owing to its military superiority, the flow characteristics of hypersonic flight is a popular area of research in many countries.

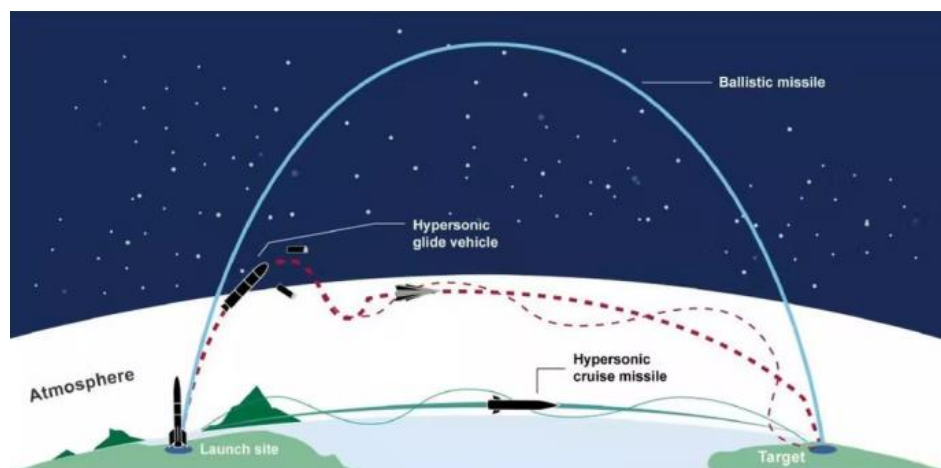


Figure 1.1 Various mission-based flight trajectories.

The major challenge in hypersonic technology is the confidentiality associated with it and the question of accuracy of the computational results as the instabilities in the numerical scheme increases. At large Mach numbers, the concept of ideal gas is no longer applicable and chemical models are required to analyse the flow characteristics. This poses another challenge in hypersonic flow as the interaction between the turbulence model and the chemical models remains poorly understood. The HIFiRE (Hypersonic International Flight Research and Experimentation) programme is a hypersonic flight test programme executed by AFRL and DSTO with an objective 'to explore fundamental hypersonic flow

characteristics, hypersonic vehicle aerodynamics and supersonic combustion performance. This program consists of 9 different geometries as shown in figure 1.2 in which the 9th geometry of HIFiRE program is currently under research. HIFiRE-1 has a simple geometry compared to other geometries and has a vast scope for numerical studies, geometry optimization for reduced heat transfer, etc.

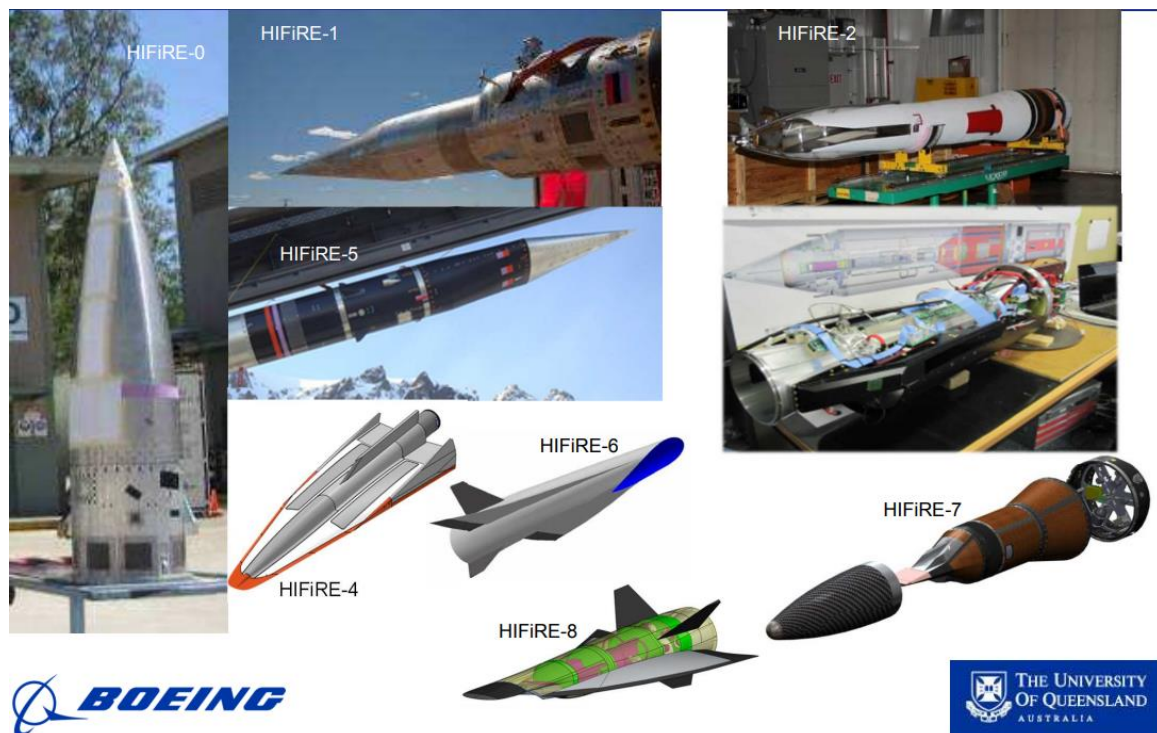


Figure 1.2 Evolution of geometries in the HIFiRE Program

The focus of this project is HF1 (HIFIRE-1) which is dedicated to study the transition and SBLI effects in hypersonic flows. The HF1 is propelled using a 2-stage Orion-Terrier Sounding Rocket. The HF1 with the sounding rocket, i.e., the experimental flight stack-up is shown in figure 1.3.

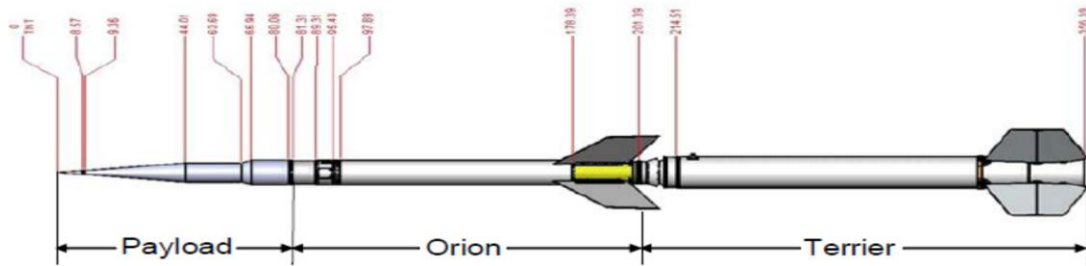


Figure 1.3 Orion-Terrier with HF1 Payload [3]

The current simulation focuses on the supersonic section of the experimental flight for reasons mentioned in the following section. Verification of the literature data provides us a foundation to build over the knowledge accumulated for venturing into the hypersonic portion of the flight.

A sustained, stable flight in such conditions requires clear understanding of the behaviour of shock-related features at different physical conditions and thereby designing of a geometry that is capable of operating in such regimes would prove to be very efficient.

2. Theoretical Background

2.1 Description of HF1 experimental payload

The payload is designed in a way as to obtain multiple flow characteristics' measurements over the same body that makes an efficient use of the experimental flight and hence, the payload is richly instrumented to measure the pressure and surface heat loads. Three experiments are conducted aboard HF1, Boundary Layer Transition (BLT), Shock Boundary Layer Interaction (SBLI) and the OMC (Optical Mass Capture) experiment.

The HF1 research vehicle incorporates a 7° half-angle cone with 2.5 mm radius. This conical section of the payload was dedicated to the primary BLT experiment. Further downstream, a 33-deg half-angle flare was designed to study the SBLI effects. This angle was determined based on the various tests performed with an aim to avail a combination of effects such as flow separation along with reattachment on the flare face. While lower angles were incapable of producing the flow separation, higher angles created large separation without any reattachment at the far end of the flare.

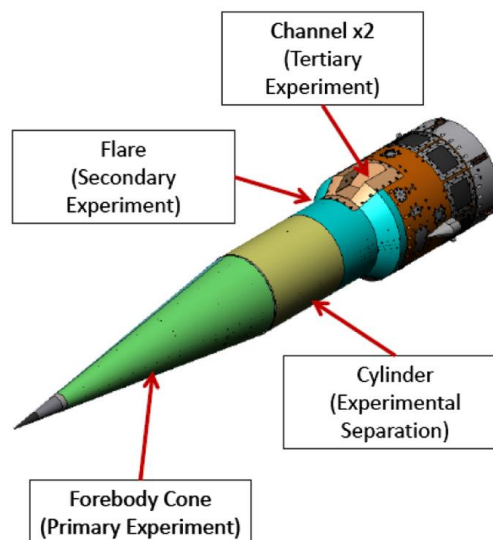


Figure 2.1 HF1 payload [11]

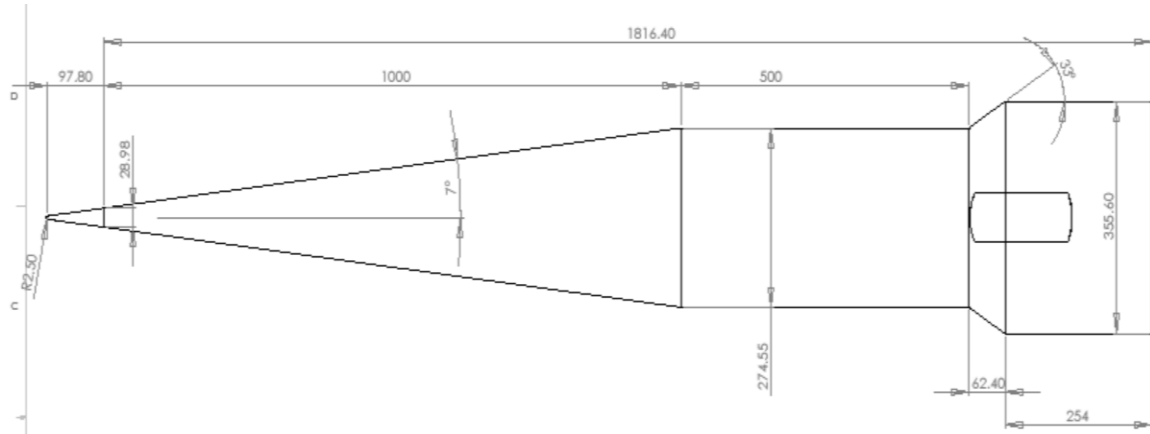


Figure 2.2 Geometrical Dimensions of HF-1 Payload [3]

The flare and the cone are separated by a cylindrical section, 0.5m long acts as an insulator in order to reduce the influence of one experiment over the other. Two channels incorporated in the flare region are dedicated to the tertiary experiment on Optical mass flux measurement, OMC [1]. The total length of the HF1 payload (test section) is 1.916m with a maximum diameter of 0.356m.

2.2 Description of HF1 Flight Test Parameters

After several ground tests at the respective facilities, an experimental flight test was carried out involving the use of an Orion-Terrier sounding rocket with the HF-1 mounted atop the stack. The Orion second stage did not separate from HF-1 during the flight as it provided the payload with passive roll stability that allowed for accurate measurements. The model used in simulations did not incorporate it for its influence upstream is considered to be minimal at such speeds [3]. The measurements of pressure and surface heat loads were made using the instrumentation installed on the vehicle. The flight trajectory is shown in Figure 2.4.

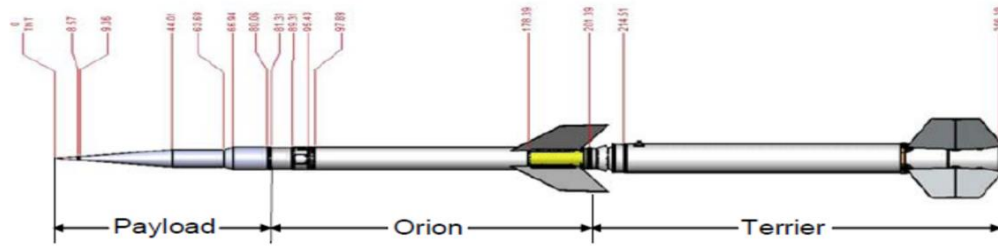


Figure 2.3 HF1 booster stack

The terrier stage fired for the first 6 seconds of the flight during which the payload is accelerated to about Mach 3. The spent stage is then drag-separated during which the payload coasts. At $t = 15$ s, the Orion stage is ignited which further accelerates the payload to Mach numbers as high as 7.8.

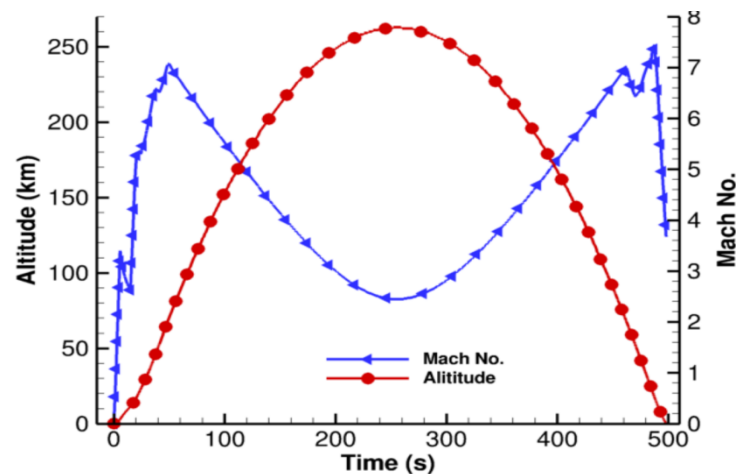


Figure 2.4 Mach no. and Altitude variation with time [10]

Measurements made at $t = 15$ s during the ascent phase is chosen as the baseline for the simulations as the Mach number is low enough for equilibrium conditions to prevail, but high enough that the aerodynamic phenomena of interest may be considered fully developed. It is also to be noted that at this point of time, the flight is at the end of its coasting phase before the ignition of the second stage where an assumption of low

vibrational load can be made. This was intended as a lower ‘noise’ environment which better highlights the subtleties of the measurement data [10]. Also, the Mach number is low enough that chemistry effects will likely be negligible but high enough that the aerodynamic phenomena expected for study may be considered fully developed. The freestream conditions associated with the simulation are mentioned in Table 2.1. In order to better replicate the experimental results in the viscous simulations, discrete values of wall temperature (Figure 2.5) distributed along the wall along with its material property is taken into consideration. This inclusion helps in better predicting the data related to heat transfer.

Table.2.1 Freestream conditions for HF1 simulation

Condition	Value	Units
Time, t	15.0	s
Altitude	11.4	km
Ma	2.63	-
P	23300	Pa
T	226	K
Re_u	$1.93e+7$	1/m

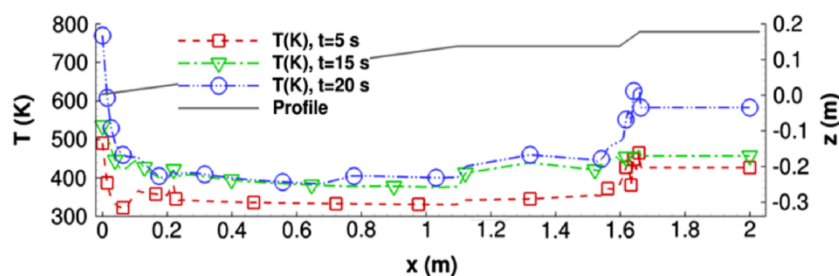


Figure 2.5 One-dimensional surface temperatures obtained from the experiment [11]

The HF-1 forecone is made up of 20mm thick aluminium shell. The nose tip is made up of a solid piece of TZM alloy, 100mm long. The coating of 50 μ m on the TZM nose acts as an oxidation barrier. The fore portion of the cylinder is an aluminium shell with thickness 5mm. The aft portion of the cylinder and flare are made up of steel. The thickness of flare wall is 12mm. The nose is connected to the cylinder with a 100mm long steel frustum and 20mm long joiner frustum [5].

2.3 Turbulence Model and Numerical Methods

The simulations are invoked with the use of RANS solver as the computational costs and resources involved in Large Eddy Simulation (LES) and Direct Numerical simulations (DNS) are quite high. This curtails the ability to predict the transition to turbulence and hence, external modifications such as specifying the transition location are necessary for obtaining solutions that perfectly agree with the experimental observations. However, determining the transition location is out of scope of our project as we focus more on the Shock Wave Boundary Layer Interactions (SBLI). For the current supersonic flow condition, a RANS based Density Solver is utilised as the flow is completely compressible in its behaviour. The AUSM scheme is chosen as the numerical method for computation of the fluid phenomena. The AUSM scheme is chosen as it is well suited for solving hyperbolic equations as well as its ability to accurately simulate the flow where discontinuities such as shock waves are present in the flow field.

Amidst the variety of turbulence models available, $k-\omega$ is chosen for its ability to predict separation effects caused by adverse pressure gradients as encountered in the flare region over the HIFiRE-1 geometry. Menter's Shear Stress Transport (SST) turbulence model with compressibility corrections is used for viscous simulations. Ideal gas fluid model is assumed to prevail as the temperatures encountered in the flow field is predominantly less than 500K, eliminating the need for inclusion of chemical models. A

brief description of k - ω model is given below. The two equations used to close the RANS equations are the length scale l_t and the velocity scale v_t equations given by

$$l_t = \frac{\sqrt{K}}{\omega} \quad \text{and} \quad v_t = \sqrt{K}$$

where ω is the energy dissipation rate per unit volume per unit time and K is turbulent kinetic energy given by

$$K = \frac{1}{2} \overline{u_i'' u_j''} = \frac{1}{2} \frac{\overline{\rho u_i'' u_j''}}{\bar{\rho}}$$

The Reynolds Stresses are related to the strain rate by the eddy viscosity (μ_t) Boussinesq approximation given by the below equation

$$-\rho \overline{u_i'' u_j''} = \mu_t \left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial \tilde{u}_k}{\partial x_k} \right) - \frac{2}{3} \delta_{ij} \bar{\rho} K$$

Out of μ_t , v_t and l_t , any two must be specified as the inflow boundary conditions. Here we input this data with the use of turbulent intensity which is considered to be 5% and turbulent to laminar viscosity ratio which is set to 0.1. The turbulent Prandtl number is set to 0.9. These values were obtained from the literature [11]. The flow numerical method is set to an upwind scheme and the viscosity model chosen is Sutherland's viscosity model. For inviscid simulations, the inviscid model is used. An axis-symmetric domain is considered for both viscous and inviscid simulations so as to simplify the computational load which additionally allows us to compare the results obtained from such a setup with those obtained from 3D simulation as well as the experimental measurements.

3. Aim and Objectives

- **Title**
Numerical Investigation of High-Speed Flow over HIFiRE-1 Flight Research Vehicle
- **Aim**
To predict the C_p distribution, skin friction coefficient and heat transfer data over the body
- **Objectives**
 1. To model the inviscid and viscous flow over the axisymmetric payload
 2. To perform a grid convergence study
 3. To compare the inviscid and viscous flow results with data from the literature
 4. To obtain an improved prediction model
- **Methods and Methodology**
The simulations were carried out in SU2 and ANSYS with the following setup.

Table 3.1 Computational Set-Up

Solver Type	RANS Density Based Solver
Fluid Model	Ideal Gas
Viscosity Model	Sutherland Viscosity Model
Convective Flow Numerical Method	AUSMPLUSUP
Turbulent model	2006 Wilcox's $k-\omega$ model
Turbulent Intensity	5%
Turbulent to Laminar Viscosity Ratio	0.1
Prandtl number	0.9
CFL number	0.1-0.55

4. Software and Simulation Set-up

The entire procedure of CFD analysis of HF1 comprises of mesh generation which involves the discretisation of the flow domain into several cells with finite volumes of their own. The solvers such as SU2 or ANSYS Fluent, work on the Finite Volume Method (FVM), a numerical method for solving PDEs that calculates the values of the conserved variables averaged across the volume. Once a converged solution is obtained, post-processing is performed for data visualisation.

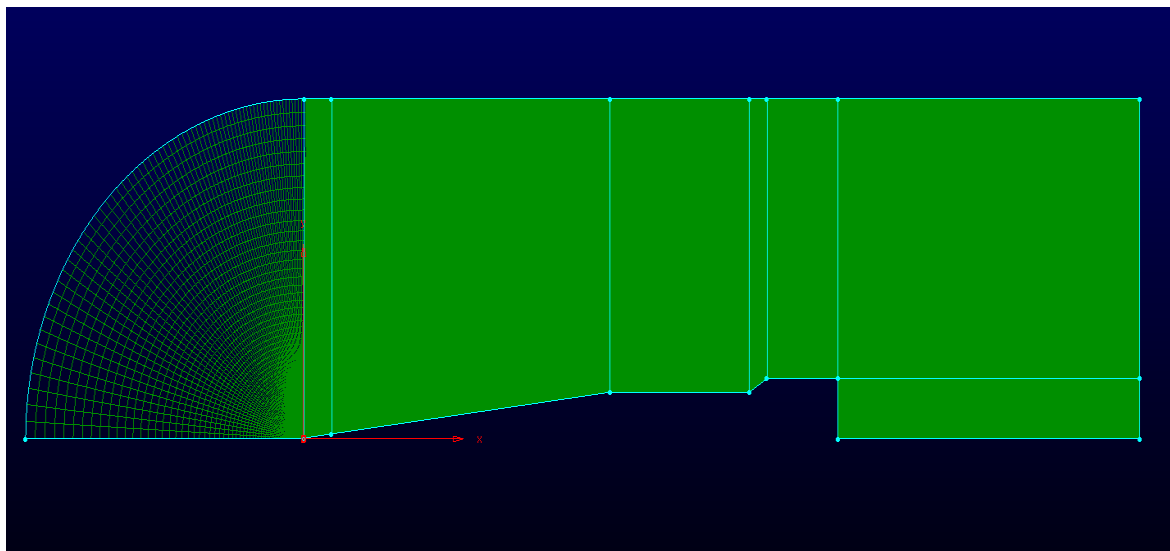
4.1 Software

SU2, an abbreviation for Stanford University Unstructured, is an open source, finite volume CFD solver. It is written using the C++ programming language and has supporting Python scripts that allow for automation of various processes and improve the ease of use. The main feature of SU2 architecture is extensibility, which can be used to treat the partial differential equations which were not envisioned previously [9]. It is due to the common framework adopted, which enables the implementation of new physics packages. These packages can be coupled to form a powerful ensemble of analysis tools which can be used to solve the complex problems faced by engineers. It provides a high-fidelity predictive capability and tools which can also be used for optimization. ANSYS Fluent is a widely used commercial CFD software based on the FVM. It can be used to model fluid flow, heat and mass transfer, chemical reactions, etc. Fluent is known for its advanced physics modelling capabilities, which include turbulence modelling, single and multiphase flows, combustion, battery modelling, fluid-structure interaction etc.

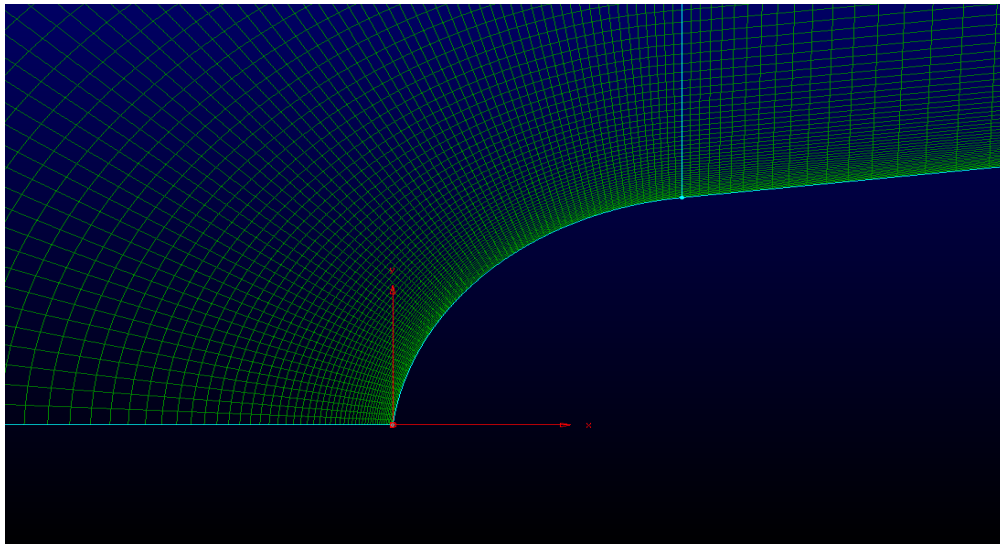
4.2 Computational Grid and Boundary Conditions

The discretization of the entire axis-symmetric flow domain was performed using the software, Pointwise. An axisymmetric grid approach was used in order to simplify the computational load. Additionally, this allows us to understand the differences in solution obtained on following such a method in comparison to 3-dimensional study. A structured grid with grid density concentrated close to the wall region was designed to obtain accurate results of surface flow related parameters such as those seen in velocity and thermal boundary layer.

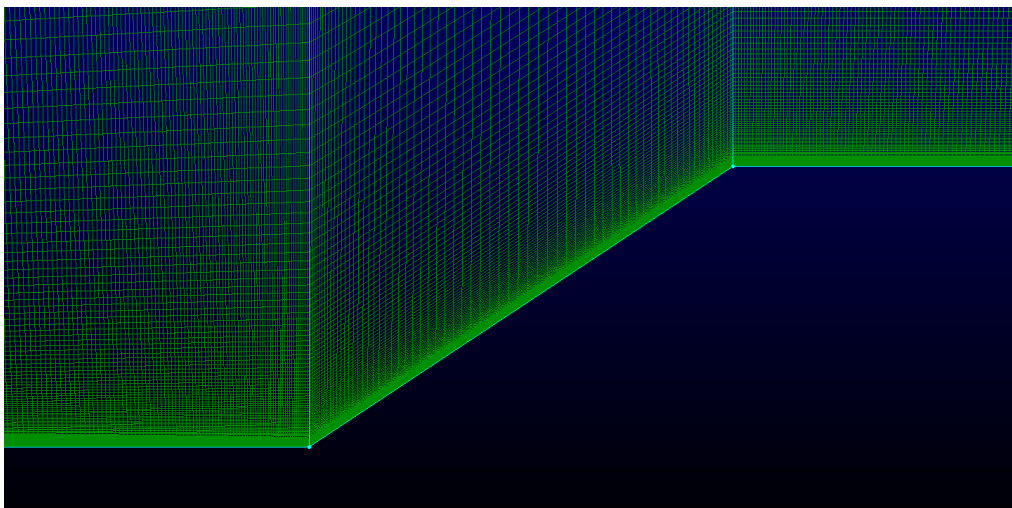
No-slip boundary condition is applied along the HF1's body wall along with specification of the wall temperature. The forward portion of the grid utilized a quarter-circle profile to mimic the 2.5mm cone at the nose-tip at the expected stagnation point. Pressure farfield boundary condition was utilised for the outer edges of the domain with flow values specified as mentioned in Table 2.1.



a) Axis-symmetric flow domain used for simulation



b) Grid distribution at the nose section



c) Grid distribution at the flare section

Figure 4.1(a-c) HF1 grid features

The mesh is locally refined to smoothly capture all geometry details and expected flow features, including boundary layers and shock waves at the nose tip and at the flare. In order to determine the best grid i.e., a grid that is able to resolve all the wall related quantities accurately, three different variants of the grid were developed with decreasing near wall spacing as reported on table 4.1.

Table 4.1 Number of cells

<i>First wall spacing</i>	<i>Number of cells</i>
1e-5	4,16,818
1e-6	5,53,024
1e-7	6,89,230

5. Results and Discussion

Inviscid and viscous simulations were performed, compared and analysed. The values of pressure coefficient, skin-friction coefficient and heat flux were determined and compared with the experimental data. The results obtained from two mesh solvers SU2 and Fluent were narrowed down to one after the comparison of C_p distribution, predicting a better model for further simulations.

5.1 Inviscid flow

Shockwaves at the nose-tip and flare region and expansion at the junction between the cone-cylinder are the primary flow features observed in the inviscid simulation.

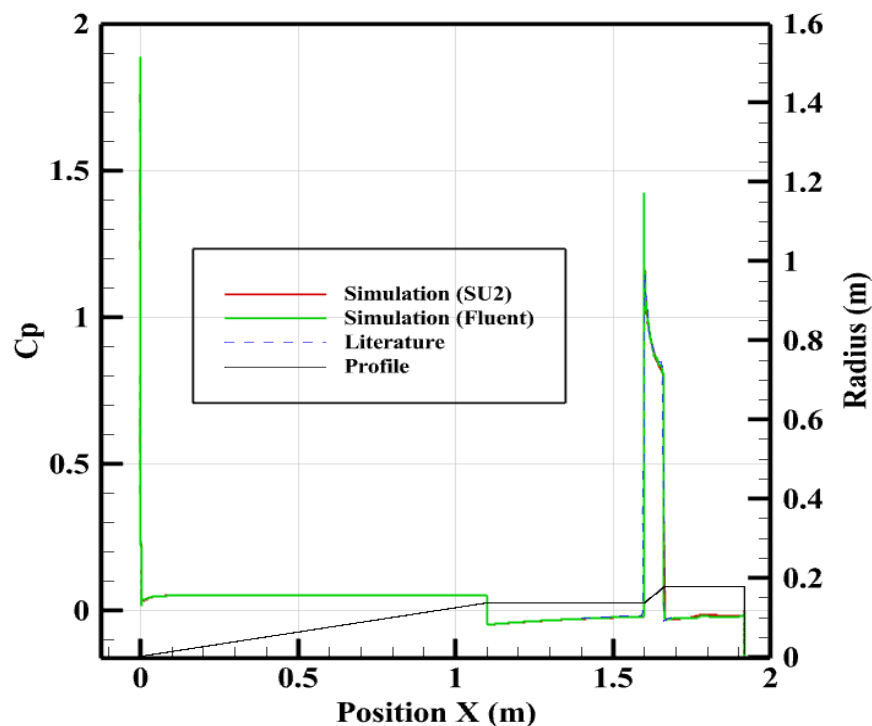


Figure 5.1 C_p distribution along the length of HF-1 under inviscid conditions.

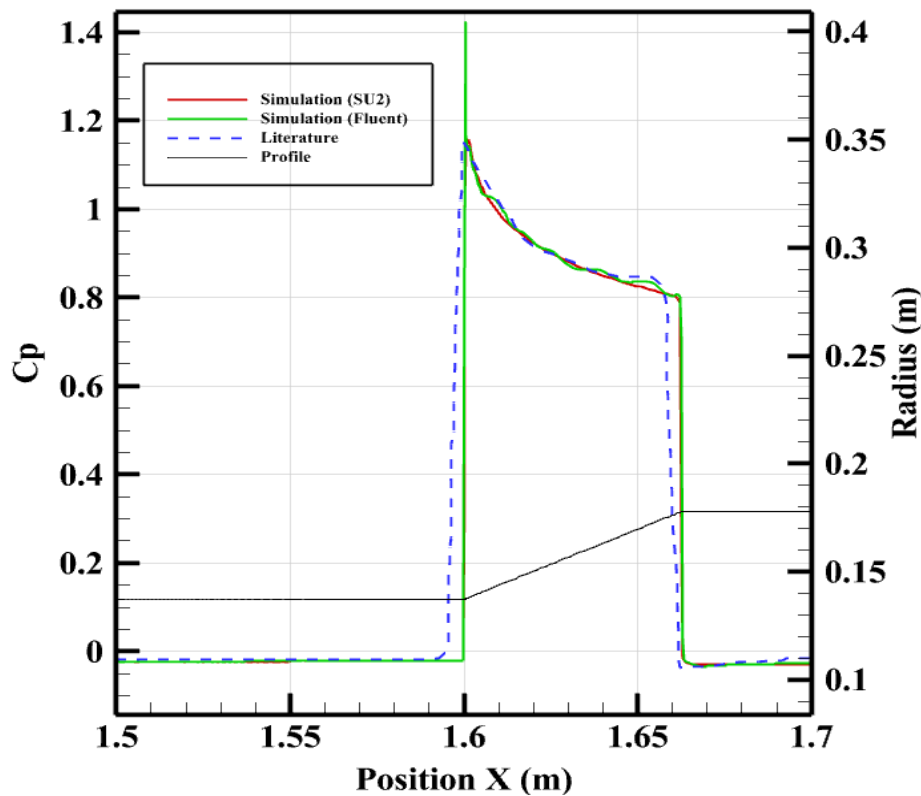


Figure 5.2 C_p distribution at flare section of HF-1 under inviscid flow conditions.

The configuration file used for inviscid flow employs AUSMPLUSUP as convective flow numerical scheme. As uniform inviscid flow approaches the nose section with large kinetic energy, a bow shock formed close to the stagnation point decelerates it with subsequent increase in pressure. Travelling downstream, the flow expands at the cone-cylinder junction followed by striking at the bottom of the flare, producing a large pressure rise at the bottom and a shock wave that causes a more uniform pressure rise on the rest of the flare which is clearly depicted in the Mach contour (Fig 5.3). Figure 5.1 also displays this sharp rise in pressure at the nose and the flare sections. Since inviscid flow doesn't account for any viscous effects, no flow separation is observed at the flare region which is confirmed in figure 5.3. The maximum value of pressure coefficient ($C_p = 1.96$) is recorded at the nose tip, that is at the stagnation point. The second highest value of C_p is

observed at the bottom of the flare due to the presence of an oblique shock wave. At the end of flare, an expansion wave results in drop of C_p values as shown in Fig 5.2. It is observed that the values of C_p at the flare section obtained from SU2 and ANSYS Fluent are in close agreement with each other and even with the literature simulation.

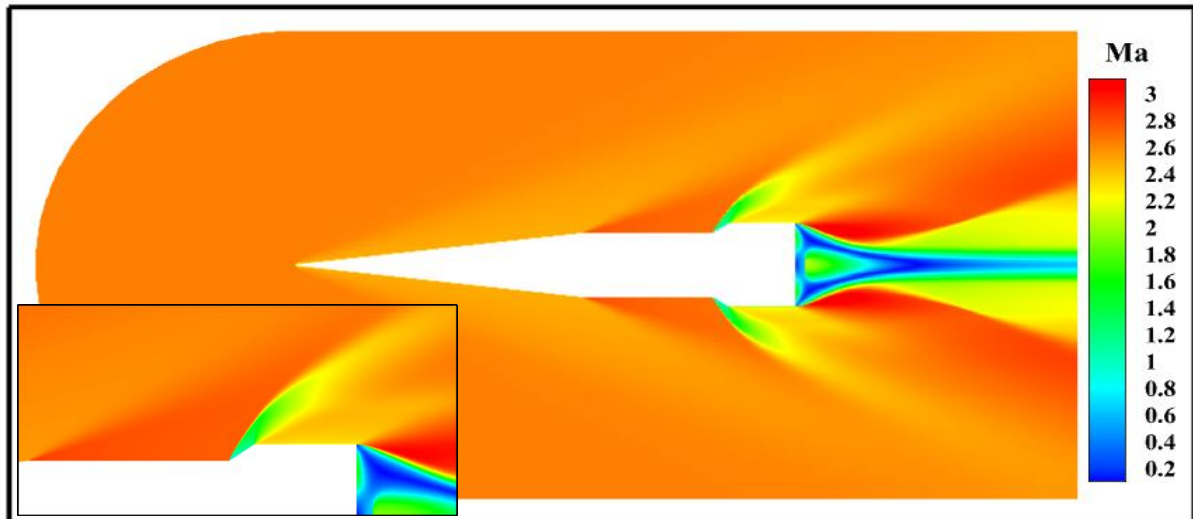


Figure 5.3 Inviscid flow Mach contour with zoomed image near flare

5.2 Viscous flow

An inviscid simulation acts as a baseline/reference for viscous simulations. Viscous flows take into account the effect of viscosity of fluid and even the wall friction, that is the influence of boundary layer on the flow quantities near the wall is considered. A major difference is observed between inviscid and viscous at the flare region. In viscous flow, there exists a high strength shock at the foot of the flare whereas in viscous flow (in Fig 5.4) Shock Boundary Layer Interactions (SBLI) occur at the flare section. The turbulent intensity is maintained and the ratio of eddy viscosity to laminar viscosity ratio are maintained at 5% and 0.1 respectively. Fig 5.5 shows a clear picture of Mach contour with streamlines and zoomed sections. At the nose section, a bow shock is observed as expected under supersonic flight conditions. A rich set of flow features are observed on the flare due to viscous effects distinct from inviscid results.

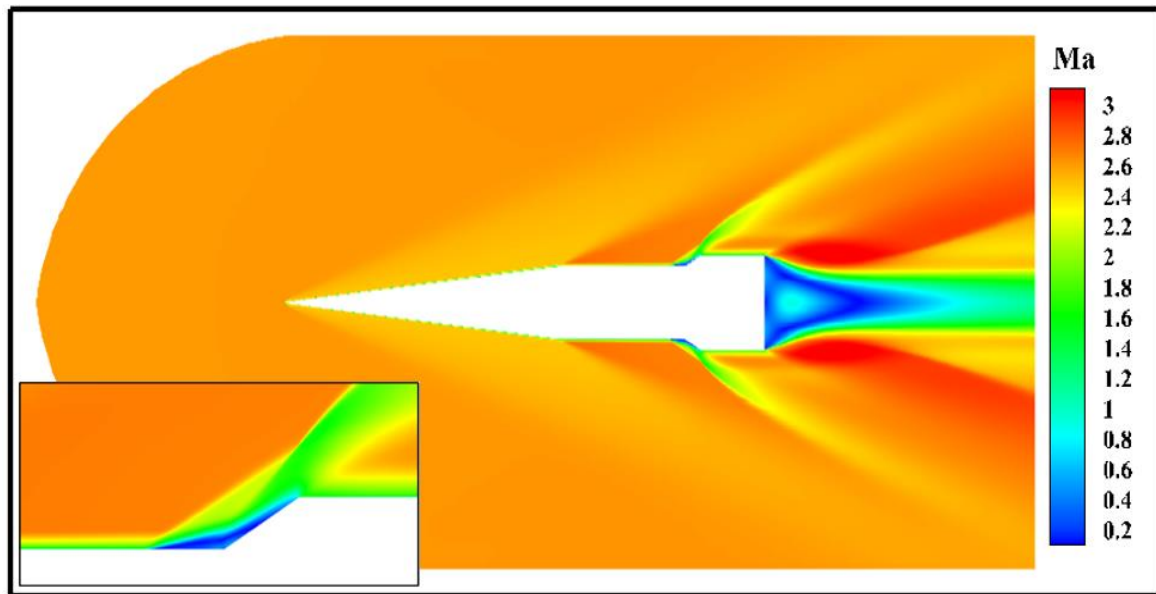


Figure 5.4 Viscous flow Mach contour with zoomed image near flare

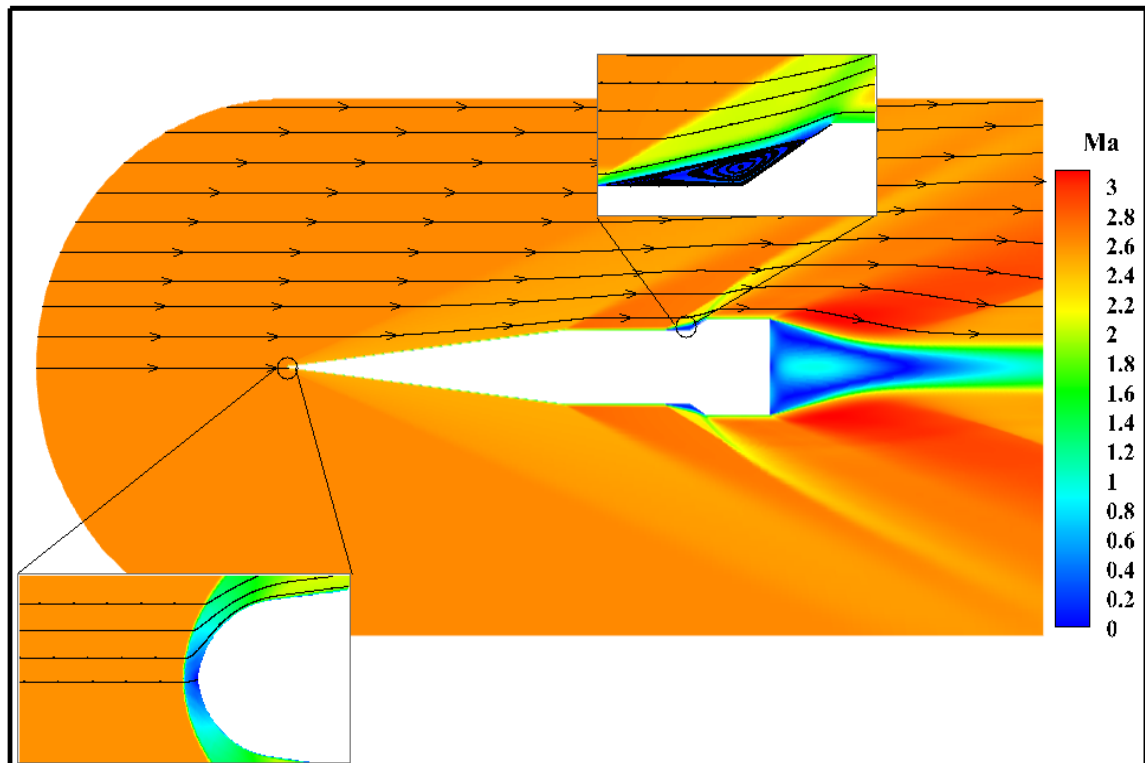


Figure 5.5 Viscous flow Mach contour with streamlines (streamlines at nose and at the flare section is highlighted)

The pressure remains constant in the fore section of the cylinder followed by a gradual increase in pressure in the aft cylinder and the flare region in contrast to sudden increase as observed in the inviscid simulations. The pressure drop in viscous flow is less than that of inviscid in the flare region as a result of viscous-effects induced flow separation. The presence of the subsonic region near the surface in the boundary layer creates substantial upstream influence in the viscous simulations. This causes separation of the boundary layer and associated SBLI as the separated region interacts with the supersonic freestream. Figure 5.5 also shows an expanded view in the vicinity of the flare and highlights the separation of the boundary layer. A shear layer over the separation bubble is observed as well as a clearly defined separation shock. The reattachment of shear layer at the end of flare gives rise to a reattachment shock wave, followed by an expansion wave. Also, a weaker third shock wave is present which is aligned with the base of the flare.

The aerodynamics of any object primarily deals with the pressure distribution over the same, as this leads to the determination of lift and drag coefficient values. The pressure distribution over HF-1 for a real flow at $M=2.63$ is interpreted in terms of pressure coefficient C_p . Figure 5.6 gives the C_p distribution along the length of HF-1. The maximum value of C_p occurs at the nose tip of the research vehicle, i.e. at the stagnation point. Until the flare section, change in values of pressure coefficient are less, however, significant variation occurs near the flare section. As seen from Fig 5.7, unlike inviscid flow, there is a gradual increase in pressure from the point of separation to the flare section. The second highest surface pressures in this simulation occur on the flare when the shear layer reattaches to the flare section.

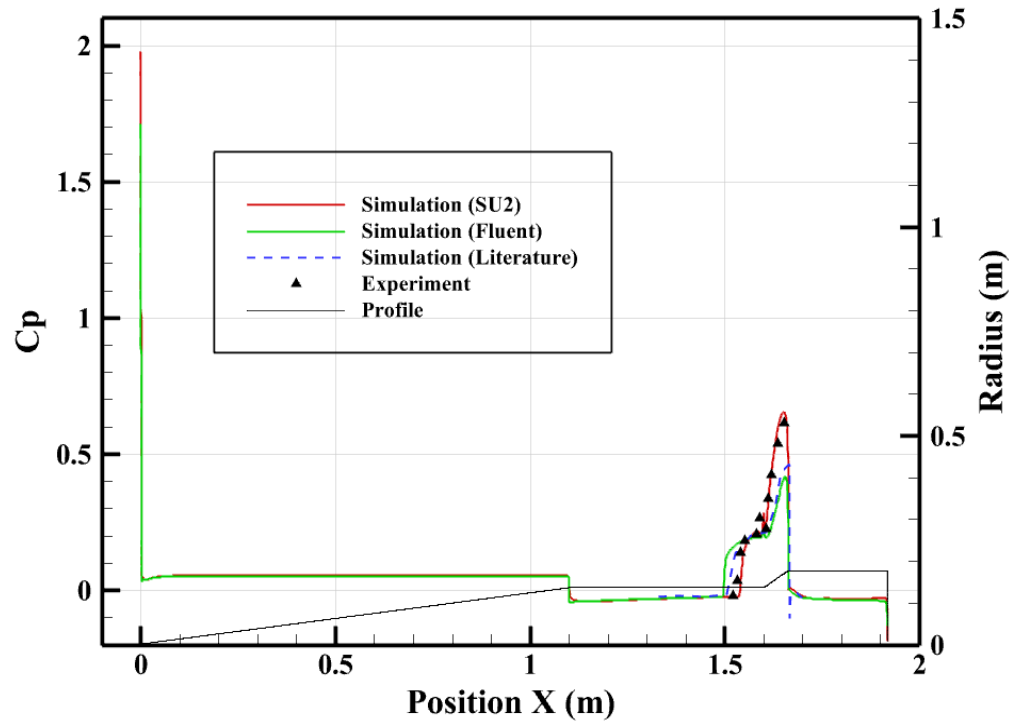


Figure 5.6 Overall C_p distribution under viscous conditions.

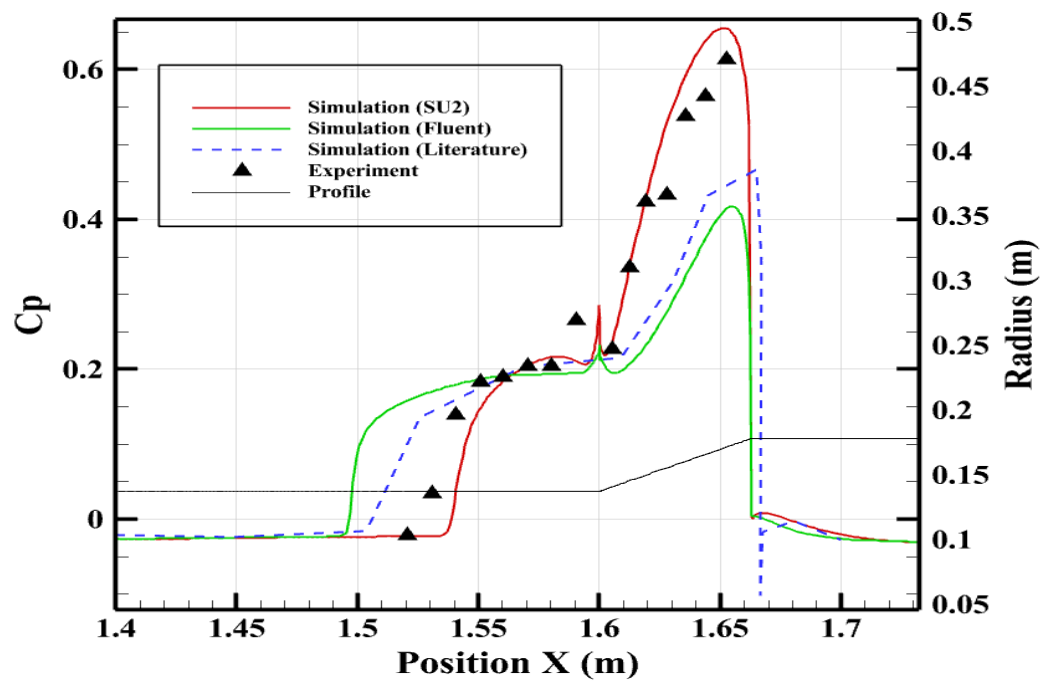


Figure 5.7 C_p distribution around the flare region under viscous flow conditions.

The comparison of results obtained from the two mesh solvers and literature simulations at the flare region is shown in figure 5.7. It can be observed that the results obtained from SU2 are in close agreement with the experiment C_p data and the results of C_p obtained from ANSYS Fluent are close to the literature simulation. The increase in pressure predicted by ANSYS Fluent is less than the actual experimental values. However, the trend of the pressure variation at the flare region remains the same in every simulation.

5.3 Comparison of Inviscid and Viscous C_p Distribution

The comparison of inviscid and viscous flow pressure distribution is carried out as shown in figure 5.8. Upper half of the contour is inviscid and bottom half of the contour is viscous. As discussed above the striking difference between the two is observed in the flare region which is distinctly visible in figure 5.8. The shock in the foot of the flare in inviscid simulation is replaced by shear layer and separation shock having considerable upstream influence in the viscous simulations. Owing to higher shock strength in the inviscid simulation, C_p is more in the inviscid simulation compared to its viscous counterpart as seen in figure 5.7. The similarity between the two simulations lies in the stagnation point where the C_p value is maximum (energy conservation principle).

Pressure distribution over HF-1 not only describes the way in which pressure is distributed but also helps in determining the lift and drag coefficient values. Since the numerical investigation is carried out on the payload of HIFiRE-1 missile, lift coefficient values are not as important as drag coefficient. On comparing the drag coefficient values given in table 5.1, the drag observed in inviscid flow is more than that in viscous flows contrary to what was expected. This is because of higher shock strength in inviscid flow than that in viscous flows. The total drag in any flow is the summation of friction drag and pressure drag (in this case wave drag). Due to flow separation at the flare section in viscous flow, local shock angle of the shock wave decreases. When compared to the shock wave created at the flare's bottom under inviscid conditions, the shock angle of the

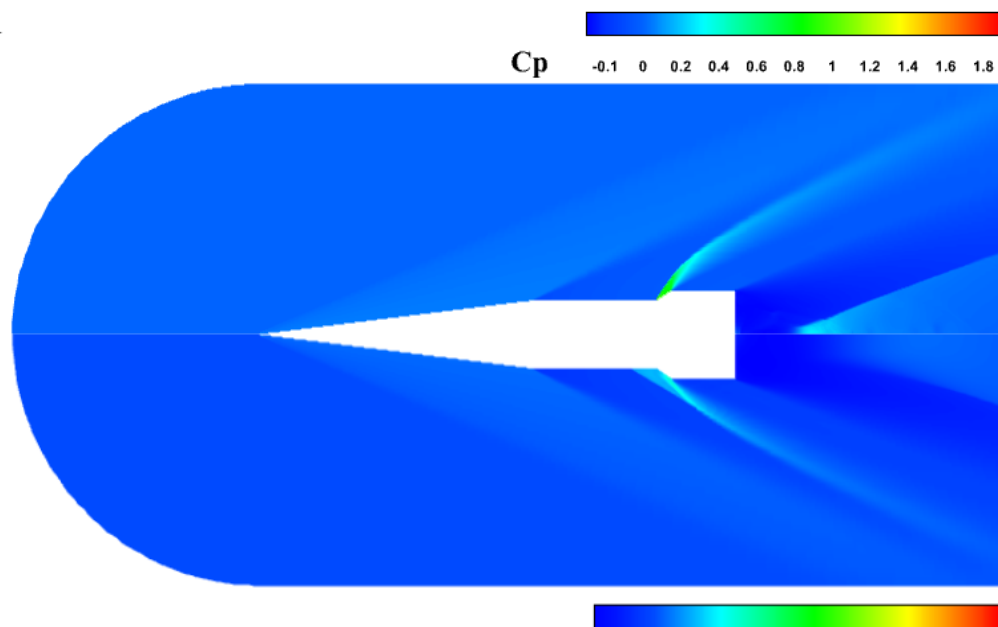
separation shock at the separation point is considerably smaller. Due to smaller shock angle, the shock strength decreases thereby results in decrease of wave drag. It can be seen that wave drag is more dominant than friction drag.

$$\text{Total drag} = \text{friction drag} + \text{wave drag}$$

In terms of drag coefficients, we have it as

$$C_{D,t} = C_{D,f} + C_{D,w}$$

Inviscid



Viscous

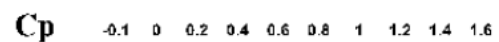


Figure 5.8 C_p contour of HF-1 inviscid vs viscous simulation.

Table 5.1 Drag coefficient values for inviscid and viscous conditions

Parameter	Inviscid	Viscous
C_D	0.531819	0.394422

Thus, the total drag over HF1 observed in inviscid condition is much higher in contrast to its viscous counterpart. Greater shock-strength in inviscid flow conditions can be attributed to causing a greater wave drag.

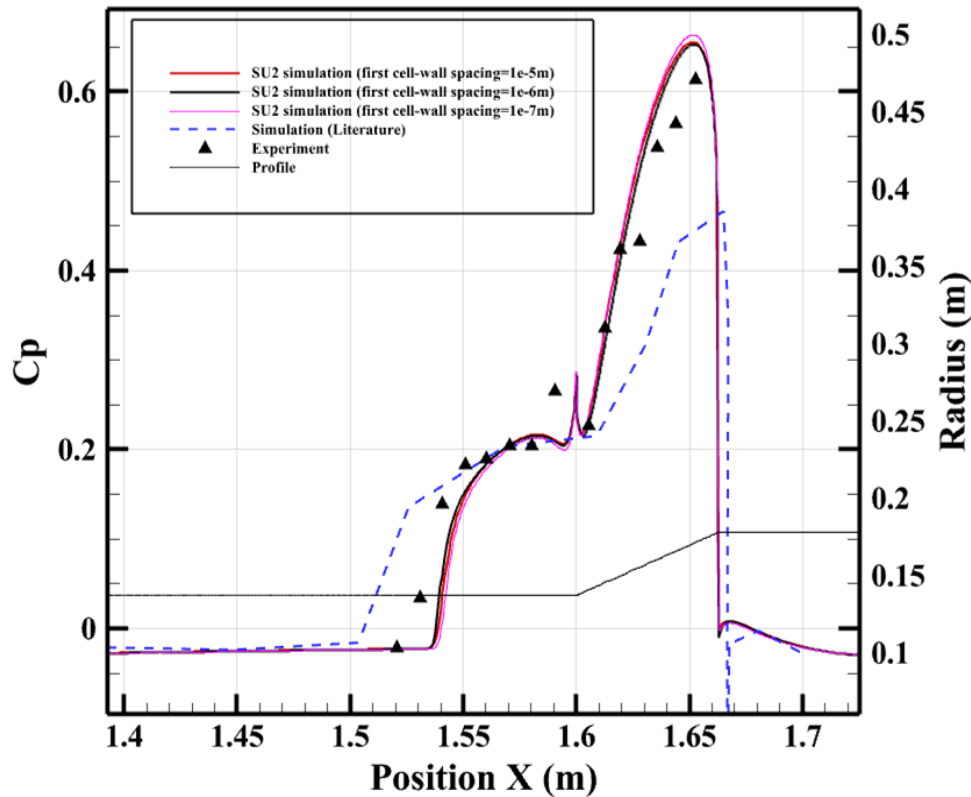


Figure 5.9 Comparison of C_p data for different grids

As part of the grid study, the comparison of three different first cell height resolution yielded curves that coincided to a great extent. Hence, the quantity c_p alone cannot be used to differentiate the effectiveness of a particular grid.

This implies that a grid with higher first-cell wall spacing will provide a sufficiently good distribution of pressure. Different grids give the same trend of C_p distribution because the changes in pressure within the boundary layer are very less compared to changes in other flow quantities. In actuality, the pressure on a wall's surface is almost identical to that at the edge of boundary layer.

5.4 Skin Friction Coefficient

The drag acting on HF-1 is not just due to pressure difference but even due to the shear stresses acting on it. Skin friction coefficient helps in easier interpretation of the friction drag (drag due to shear stresses). This coefficient specifies the fraction of dynamic pressure, that is felt as shear stress on the surface. It is given by the below formula which shows that it depends on shear stress, free stream density and velocity.

$$\text{Skin Friction Coefficient, } C_f = \frac{\tau_w}{\frac{1}{2}\rho u^2}$$

Where τ_w is the wall shear stress.

The skin friction coefficient in stagnation region is more due to high shear wall stress (Fig 5.10). There is a local rise in C_f at the nose cylinder interface due to the presence of expansion waves and it remains constant till the point of flow separation. The negative sign of C_f in the separation region as shown in Fig 5.11 can be attributed to the flow reversal. The steep increase in C_f in the aft flare region can be due to the reattachment of shear layer. It is interesting to note that the trends observed in C_p and C_f plots are the same except in the separation region owing to flow reversal.

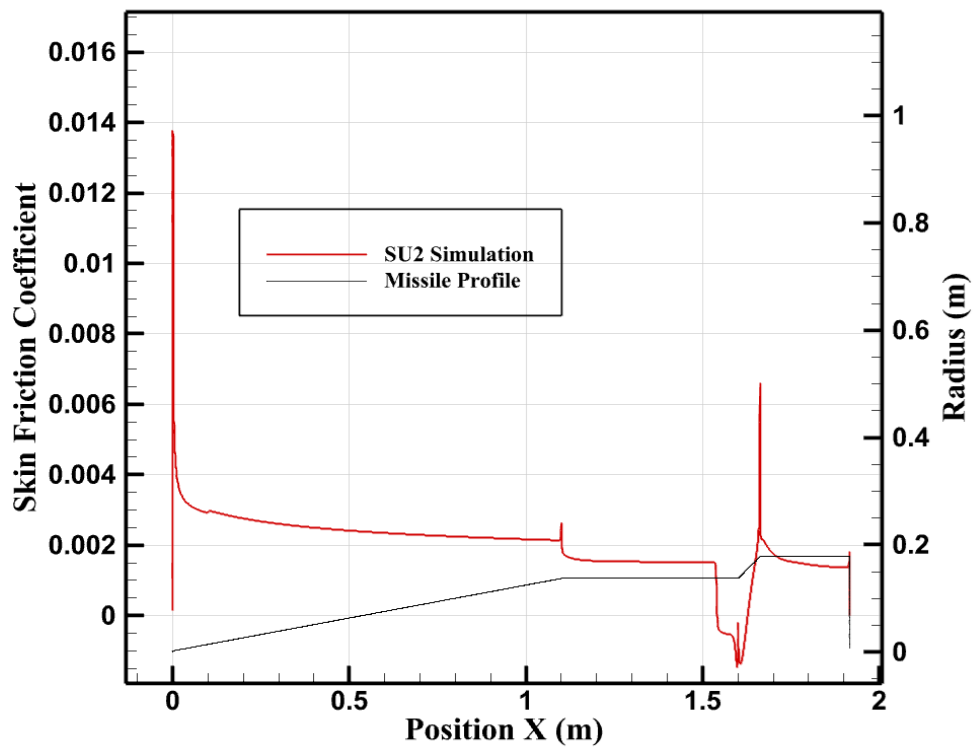


Figure 5.10 Overall C_f distribution of HF-1.

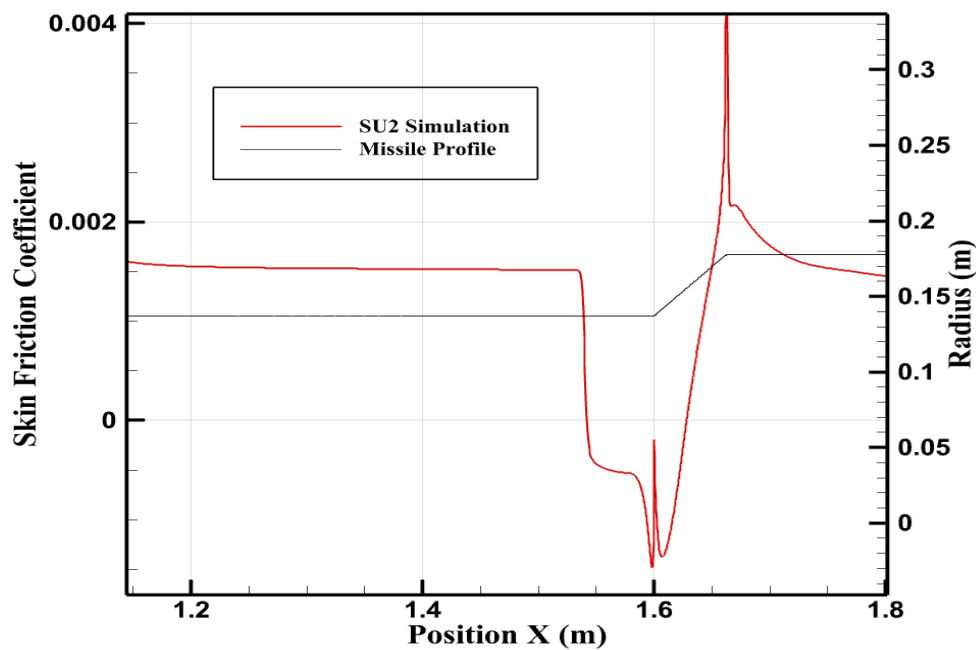


Figure 5.11 C_f distribution around the flare region.

5.5 Heat Flux

In high-speed flows, in addition to the pressure distribution over an aerodynamic body, heat transfer to the body becomes more significant and inevitable to estimate, as heat transfer rates are quite large at high Mach numbers. If this heat transfer is uncontrolled, it can lead to material damage, failure of engine and flight vehicles. Selection of appropriate materials for external skin of vehicles (mainly the front section) is necessary to sustain heat transfer at large Mach numbers. Heat flux is defined as heat transfer rate per unit area. Its unit is W/m^2 . Thus, estimating heat flux is important yet difficult to obtain accurate values with computational techniques. The accuracy of heat flux values depends on the refinement of grid at the wall section and also on the first node spacing from wall. A refined grid can determine better gradients of temperature $\left(\frac{\partial T}{\partial y}\right)$ along the normal to wall than a coarse grid.

Heat flux can be handled better and interpretation becomes easier in terms of a dimensionless quantity. This dimensionless quantity is known as ‘Stanton Number’. Stanton number is defined as ‘the ratio of heat transferred into a fluid to the thermal capacity of fluid’. Mathematically, it is expressed as

$$C_h = \frac{\dot{q}_w}{\rho_e U_e C_p (T_{aw} - T_w)} \quad \dots (1)$$

Here, $\dot{q}_w$ is the heat flux, ρ_e is the free-stream density, U_e the free-stream velocity and C_p is the specific heat at constant pressure (J/kgK). T_{aw} represents the adiabatic wall temperature in K and T_w is the wall temperature.

In case of HIFiRE-1, the experimental data for heat transfer obtained from literature [11] is expressed in terms of Stanton number. This Stanton number is converted to heat flux through following steps:

Step 1: Obtain Total Temperature

Given the quantities of ratio of specific-heat γ , free-stream Mach number M_∞ and temperature T_∞ , through isentropic relations, total temperature T_o is determined.

$$T_o = T_\infty \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right)$$

Step 2: Calculation of Recovery Factor

The recovery factor r is calculated based on whether the flow is laminar or turbulent. The table below lists the expression for recovery factor

Table 5.2 Expressions for recovery factor

Type of flow	Recovery factor r
Laminar flow	$\sqrt{Pr}$
Turbulent flow	$(Pr)^{1/3}$

Here, Pr is the Prandtl number (ratio of momentum diffusivity to thermal diffusivity). In case of HF-1, Prandtl number is 0.9 at $t=15s$.

Step 3: Determination of Adiabatic Wall Temperature

On obtaining the recovery factor r and total temperature T_o , adiabatic wall temperature can be computed using the expression

$$T_{aw} = r(T_o - T_e) + T_e$$

where T_e is the temperature of flow at the edge of boundary layer which is equal to free-stream temperature.

Step 4: Determination of heat flux

Heat flux is obtained by substituting the values of Stanton number, free-stream density and velocity, specific heat, adiabatic wall temperature and wall temperature in equation (1) i.e.

$$\dot{q}_w = C_h \times \rho_e U_e C_p (T_{aw} - T_w) \quad \dots (2)$$

The experimental data of Stanton number at t=15s is converted to heat flux and compared with the heat flux obtained from simulations.

The heat transfer values at the wall boundary obtained are compared between grids with different first cell-wall spacings (1e-5, 1e-6 and 1e-7 m) and the experimental data obtained from literature [11]. Grids with first cell-wall spacing of 1e-6 and 1e-7m give better heat transfer results near the flare section than the grid with first cell-wall spacing of 1e-5m. From figure 5.10, it is observed that the simulations performed from SU2 have the same trendline as those obtained from the experimental flight. At the separation point, heat flux increases gradually, attains a peak value at foot of the flare and drops down at the end of flare section.

Maximum value of heat transfer occurs at the nose tip as expected. Large amount of kinetic energy is converted to internal energy increasing temperature and heat transfer at the nose region (stagnation point). The second maximum value of heat transfer occurs at the bottom of flare section shown in Fig 5.12. The heat transfer values from the point of separation to the flare's bottom can be inferred from the aforementioned plot, which shows that they are well in line with the experimental data. However, the heat transfer values at the flare section are very different from the experimental measurements. This is because the simulations ran for a smaller duration of time. If the simulations were made to run for a longer time, then heat flux values would be in good agreement with experiment at the flare region as well.

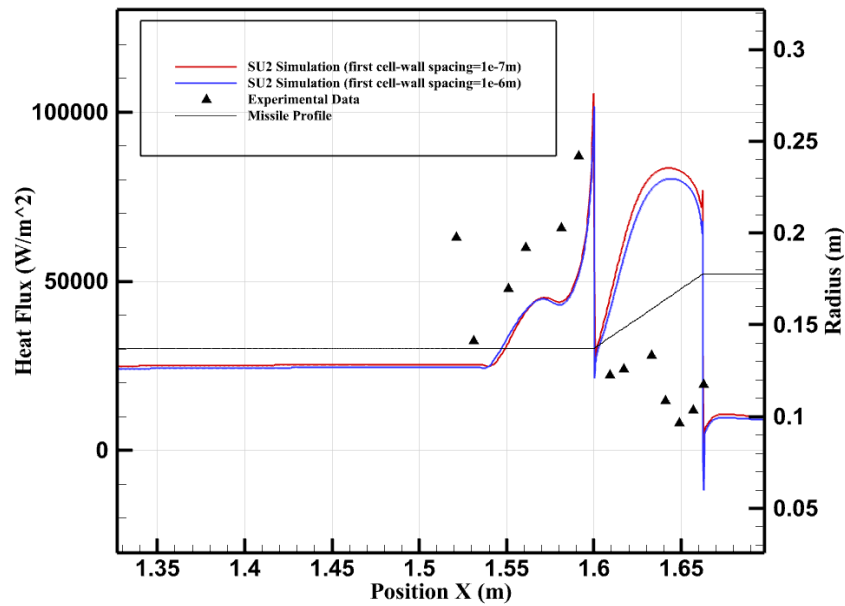


Figure 5.12 Comparison of heat flux with experimental data

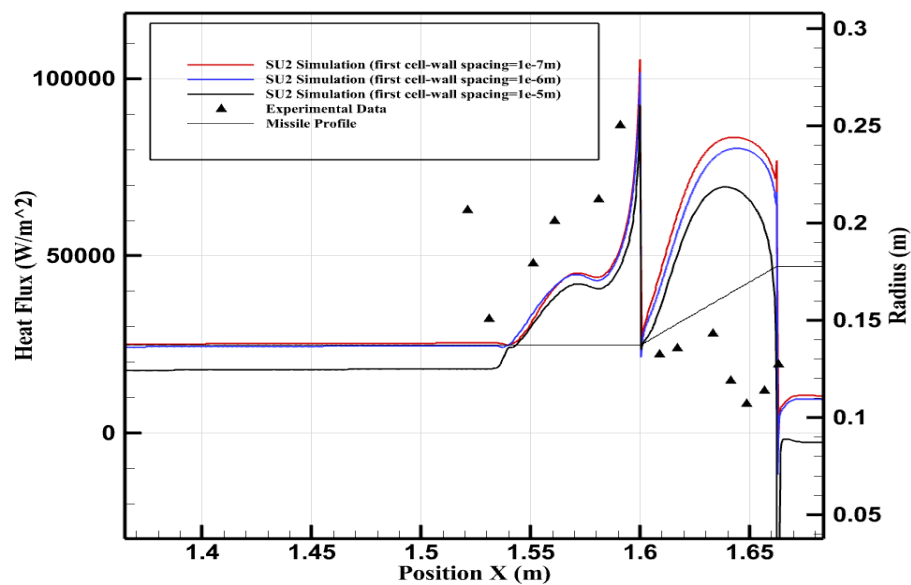


Figure 5.13 Comparison of heat flux data for different grids

On comparing the heat flux data obtained from simulations, the effect of first cell resolution can be identified which was prime goal of performing a grid study. Better resolution i.e., with lower first cell height, a better heat flux prediction is obtained which

can be observed in Fig. 5.13. This is due to accurate temperature gradient calculation close to the body wall. From the above comparison plot, the heat flux obtained from simulation agrees well with the fore section of the flare at the separation region whereas fails to accurately predict the quantity at the flare region. The cause of such difference could be attributed to errors associated with experimental flight data which studied as part of future work.

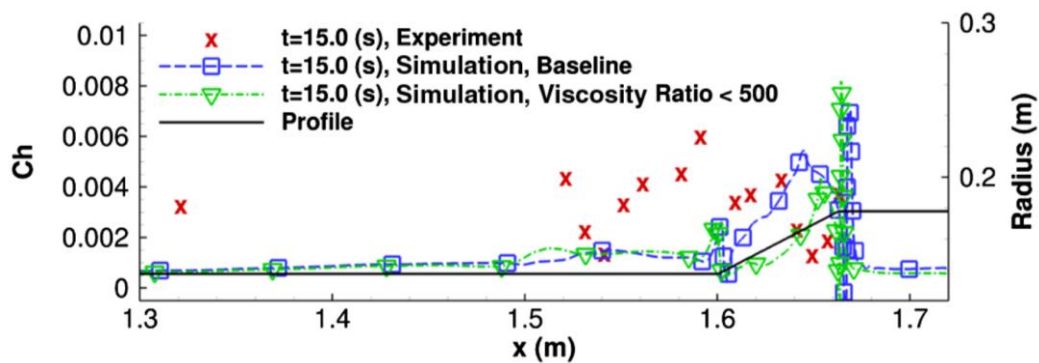


Figure 5.14 Heat transfer coefficient variation [11]

Also, to be noted here is the effectiveness of the simulations performed in this project. The data obtained from the axisymmetric simulation was able to well predict the heat flux variation at the separation region whereas the simulation performed by Kimmel et. al. [11] fails to capture the trend at the separation region.

6. Conclusions and Suggestions for Future Work

The flow over HIFiRE-1 was successfully investigated with the results obtained agreeing well with the experimental measurements. The use of Wilcox $k-\omega$ SST turbulence model proved to be the right choice for its ability to reproduce the complex flow fields at the flow separation region ahead of the flare. The advantages of using AUSM scheme as the discretising numerical method is also observed in the results obtained. The pressure coefficient, C_p distribution over HF1 obtained through simulations was very close to those from literature. For the inviscid and viscous simulations that were performed, the most striking difference was observed in the vicinity of the flare, where the viscous simulations show significant upstream influence of the flare as a result of the Shock Boundary Layer Interaction, which results in spreading of the interaction region thereby reducing the shock strength.

The simulation also provided an understanding of the heat flux component of the flow where convective effects needed close examination during setup. With a portion of the heat flux close to the flare, the separation region to be precise, matching the experimental data better than the values as seen from the literature indicates the effectiveness of the axisymmetric simulation performed. It is noticed that SU2 provided a better numerical approximation of pressure and heat flux distribution over HF-1 to the experimental values than Fluent, while that of the latter better agrees with the literature simulation results. As axisymmetric simulations do not account for the effect of the two channels at the flare, a three-dimensional simulation of HF1, once performed, may provide results that are in better agreement with the experimental data. Three-dimensional simulations with a better optimized geometry can also be considered in the future work.

References

1. Anderson, J. D., *Hypersonic and High-Temperature Gas Dynamics*, 2nd ed., AIAA Education Series, AIAA, Reston, VA, 2006, pp. 16–22, 449–785
2. Holden, M. S., Wadhams, T. P., Harvey, J. K., and Candler, G. V., “*Comparison Between Measurements in Regions of Laminar Shock Wave Boundary Layer Interactions in Hypersonic Flows with Navier–Stokes and DSMC Solutions*,” NATO Air Vehicles Technology Research and Technology Office, RTO-TR-AVT-007-V3, 2002.
3. Kimmel, R. L., and Adamczak, D., “*HIFiRE-1 Background and Lessons Learned*,” 50th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition, AIAA Paper 2012-1088, Jan. 2012, pp. 1–15
4. Kimmel, R. L., “*Aerothermal Design for the HIFiRE-1 Flight Vehicle*,” 38th Fluid Dynamics Conference and Exhibit, Seattle, WA, AIAA Paper 2008-4034, June 2008.
5. Kimmel, Roger. (2008). *Aerothermal Design for the HIFiRE Flight Vehicle*. 10.2514/6.2008-4034.
6. Kimmel, R. L., and Adamczak, D., “*HIFiRE-1 Background and Lessons Learned*,” 50th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition, AIAA Paper 2012-1088, Jan. 2012, pp. 1–15
7. Kimmel, R. L., and Adamczak, D., “*HIFiRE-1 Preliminary Aerothermodynamic Measurements*,” 41st AIAA Fluid Dynamics Conference and Exhibit 27–30 June 2011, Honolulu, HI, AIAA Paper 2011-3413, June 2011, p. 35.
8. Stanfield, S. A., Kimmel, R. L., and Adamczak, D., “*HIFiRE-1 Flight Data Analysis: Turbulent Shock/Boundary-Layer Interaction Experiment During Ascent*,” 42nd AIAA Fluid Dynamics Conference and Exhibit, New Orleans, LA, AIAA Paper 2012-2703, June 2012.

9. Thomas D. Economon., Francisco Palacios., Sean R. Copeland., Trent W. Lukaczyk., A Juan J. Alonso., *"SU2: An Open-Source Suite for Multiphysics Simulation and Design,"* AIAA Journal Vol. 54, No. 3, March 2016, pp. 828-846.
10. Yentsch, R. J., and Gaitonde, D. V., *"Numerical Investigation of Hypersonic Phenomena Encountered in HIFiRE Flight 1,"* AIAA ASM Nashville, AIAA Paper 2012-0943, Jan. 212, pp. 1–12.
11. Yentsch, Robert & Gaitonde, Datta & Kimmel, Roger. (2013). *Performance of Turbulence Modeling in Simulation of the HIFiRE-1 Flight Test.* Journal of Spacecraft and Rockets. 51. 10.2514/1.A32535.